

Physical AI SDK - Model Lifecycle & Strategy

2026

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AECG/PAVS AI SW

Model Lifecycle Overview

The Four-Stage Model Lifecycle



Key Phases:

1. **Enablement** - Initial model support, integration, basic functionality
2. **Profiling** - Performance analysis, bottleneck identification, hotspot analysis
3. **Benchmarking** - KPI validation, performance metrics, competitive analysis
4. **Fine-tuning** - Model optimization, domain-specific adaptation, deployment ready

Progressive Model Support Strategy

From Individual Models to System Orchestration

Phase 1: Individual Model Support

- Focus on single model enablement and optimization
- Establish baseline performance for each model
- Build reusable infrastructure and tooling

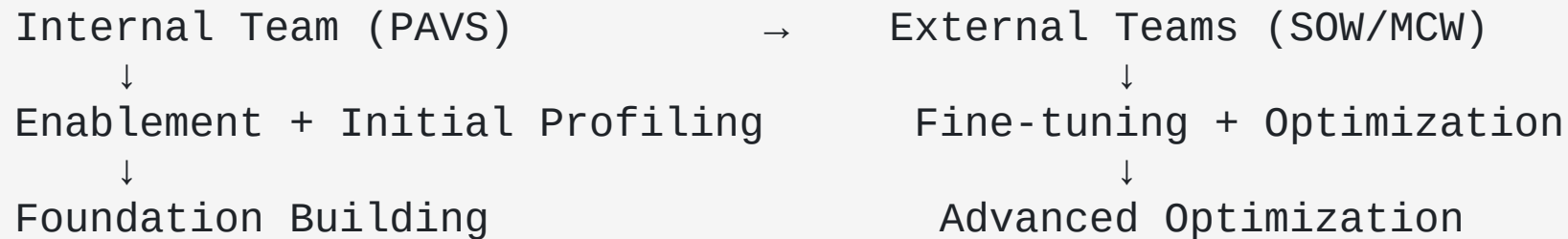
Phase 2: Multi-Model Orchestration

- Coordinate multiple models working together
- System-level optimization and resource management
- End-to-end pipeline integration

Example Pipeline: Vision → VLM → LLM → TTS

Effective Team Collaboration

Progressive Engagement Model



Strategy:

- **PAVS Team:** Handle enablement and initial profiling - establish working baseline, identify optimization opportunities
- **External Partners:** Focus on fine-tuning and advanced optimization - domain-specific optimization, large-scale benchmarking

Benefits: ✓ Better ROI ✓ Clear handoff criteria ✓ Faster time to production

Speed Matters - Parallel Execution

Current Model Status Across Lifecycle Stages

Model Name	Category	Enablement	Profiling	Benchmarking	Fine-tuning
MobileSAM	Vision/Segmentation	✓	✓	✓	🔄
YoloV12	Object Detection	✓	✓	✓	⏸
Qwen3-VL	VLM	✓	✓	🔄	⏸
LLaMA3.2-Vision	VLM	✓	✓	⏸	⏸
Whisper	ASR	✓	✓	✓	✓
XTTS	TTS	✓	✓	✓	⏸
DeepSeek-R1	LLM	✓	🔄	⏸	⏸
Qwen2.5-VL	VLM	✓	✓	✓	🔄
CrossFormer	Vision Transformer	✓	🔄	⏸	⏸
CenterPoint	3D Detection	✓	⏸	⏸	⏸
OpenVLA	Vision-Language-Action	✓	⏸	⏸	⏸
EfficientNetV2	Image Classification	✓	✓	✓	⏸

Legend: ✓ Complete | 🔄 In Progress | ⏸ Not Started

Environment & CI/CD Infrastructure

Isolated Environments for Each Model/System

Environment Strategy:

```
physical-ai-sdk/  
├── models/  
│   ├── mobilesam/  
│   │   ├── Dockerfile  
│   │   ├── pyproject.toml (uv)  
│   │   └── .venv/  
│   └── yolov12/  
│       ├── Dockerfile  
│       ├── requirements.txt  
│       └── .venv/  
└──
```

Benefits: ✓ Isolated dependencies ✓ No version conflicts ✓
Easy model-specific exploration

Branch Strategy:

- `feature/*` → Developer branches
- `dev` → **Continuous work branch**
- `prod` → **Stable release branch**

Model Deployment Flow:

1. Developer → Push to `dev`
2. CI/CD runs automated tests
3. Validated → Pull to `prod`
4. Final validation suite
5. Deploy to customers

Container & Virtual Environment Options

Deployment Flexibility

Option 1: Dockerfile

```
FROM rocm/pytorch:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install -r \
    requirements.txt
COPY . .
CMD ["python", \
    "inference.py"]
```

Option 2: uv + pyproject.toml

```
[project]
name = "mobilesam"
version = "1.0.0"
dependencies = [
    "torch>=2.0.0",
    "onnxruntime-rocm",
]
```

Option 3: pip + requirements

```
torch==2.0.0
onnxruntime-rocm==1.17.0
opencv-python==4.8.0
```

Expanding Model Portfolio

Staying Current with Latest Open Source Models

Emerging Neural Alternatives:

-  **End-to-End Neural Pipelines** replacing traditional CV pipelines
 - Example: Neural SLAM replacing traditional SLAM
 - Example: Neural 3D reconstruction replacing classical methods
-  **Latest Model Categories to Track:**
 - Multimodal Foundation Models (GPT-4V alternatives)
 - Action-VLMs (RT-2, OpenVLA successors)
 - Neural Rendering (NeRF, Gaussian Splatting for robotics)
 - On-device Edge LLMs (< 3B parameters)

Strategy:

1. Monthly scan of arxiv.org, HuggingFace, GitHub trending
2. Quick feasibility assessment (30-day cycle)
3. Add to roadmap if aligned with Physical AI use cases

Benchmarking & Documentation Strategy

GitHub Pages Integration

Documentation as Code:

```
docs/
├── index.md
├── models/
│   ├── mobilesam.md
│   ├── yolov12.md
│   └── benchmarks/
│       ├── robotics.md
│       └── industrial.md
├── guides/
│   ├── quick-start.md
│   └── deployment.md
└── _config.yml
```

Automated Workflow:

1. **Code + Benchmarks** → GitHub Repo
2. **CI/CD** → Auto-generate docs, run benchmarks
3. **GitHub Pages** → Publish results automatically

Benefits:

- ✓ Version-controlled documentation
- ✓ Automated benchmark updates
- ✓ Professional presentation
- ✓ Community collaboration

Comprehensive Benchmarking Strategy

Dual-Mode Benchmarking & Regression Testing

1. NPU-Disabled Mode (Competitive Comparison)

- **Purpose:** Fair comparison with NVIDIA/Intel
- **Hardware:** NPU disabled, iGPU + CPU execution
- **Use Case:** Match industry-standard metrics

2. NPU-Enabled Mode (AMD Advantage)

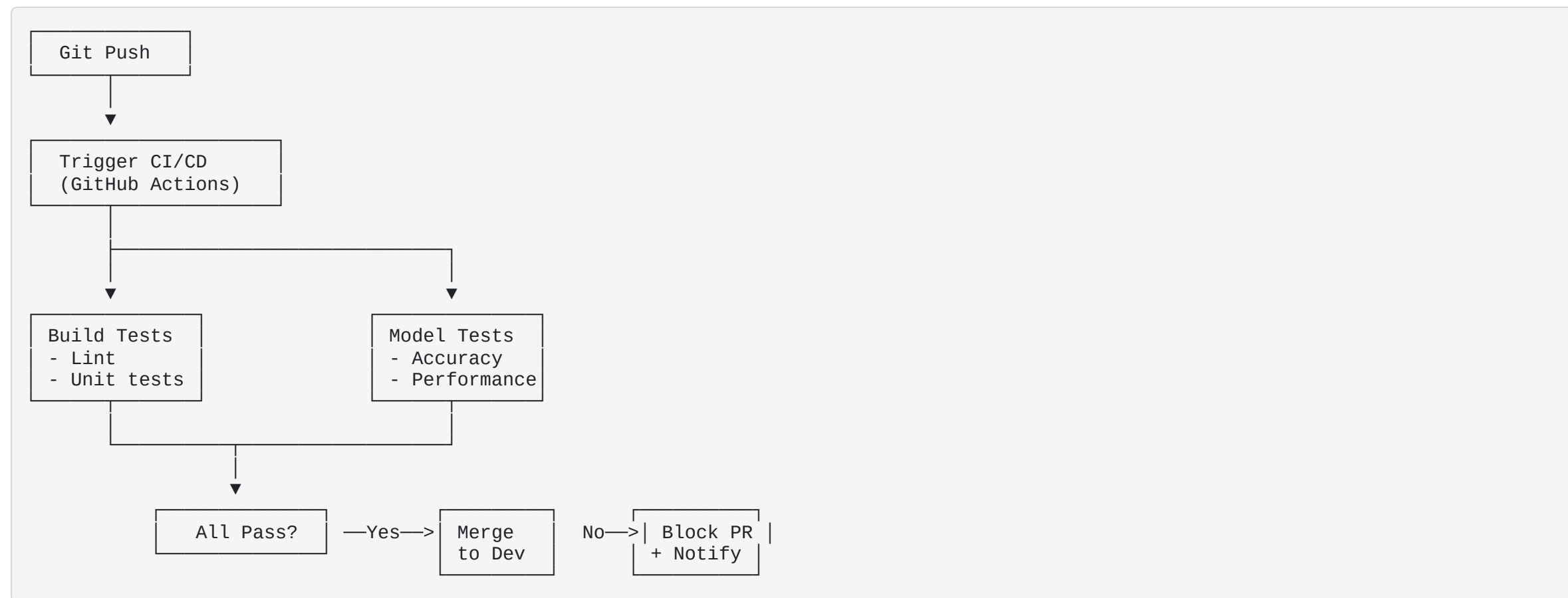
- **Purpose:** Demonstrate AMD's hybrid execution
- **Hardware:** NPU + iGPU + CPU (full APU)
- **Use Case:** Showcase AMD performance gains

Performance Comparison:

Model	NPU-Off (vs Competitors)	NPU-On (AMD Advantage)	Gain
MobileSAM	180 fps (iGPU)	240 fps (NPU)	+33%
YoloV12	200 fps (iGPU)	240 fps (NPU)	+20%
Whisper	85 tok/s (CPU)	100 tok/s (NPU)	+18%
Qwen3-VL	25 fps (iGPU)	35 fps (NPU+iGPU)	+40%

CI/CD Pipeline Detail

Preventing Regressions



Practical Example - Model Lifecycle

Case Study: Qwen3-VL Integration

Week 1-2: Enablement

-  Model weights download and conversion
-  Basic ONNX-RT integration
-  Simple inference test (single image)
-  Docker environment setup

Week 3-4: Profiling

-  Layer-wise performance analysis (AMD Profiler)
-  Memory usage tracking
-  Hotspot identification (Conv3D, GEMM)

Result: 2.5x speedup, 92% accuracy maintained

Week 5-6: Benchmarking

-  KPI testing: Target 30fps @ 1080p
-  Accuracy validation on VQA datasets
-  Power consumption measurement

Week 7-8: Fine-tuning

-  Triton kernel optimization for GEMM
-  INT8 quantization (W8A8)
-  Model-specific automotive dataset fine-tuning

Resource Management Strategy

Balancing Internal & External Resources

Resource Allocation Matrix:

Lifecycle Stage	Internal Team (PAVS)	External Partners	Duration
Enablement	Primary	Support	2-3 weeks
Profiling	Lead	Assist	2-4 weeks
Benchmarking	Guide	Execute	3-4 weeks
Fine-tuning	Review	Primary	4-8 weeks

Handoff Criteria:

- ✓ Model runs successfully on target hardware
- ✓ Profiling report completed
- ✓ Baseline benchmarks established
- ✓ Optimization opportunities documented

Technology Watchlist 2026

Emerging Models & Technologies

High Priority:

1. **Qwen3-VL 32B** - Latest multimodal foundation model
2. **RT-X** - Robotics transformer with broader action space
3. **Neural SLAM** - End-to-end learned SLAM systems
4. **Depth Anything V2** - Improved monocular depth

Medium Priority:

5. **SAM 2** - Segment Anything with video support
6. **Florence-2** - Microsoft's vision foundation model
7. **Stable Diffusion 3** - For synthetic data generation
8. **Phi-3 Vision** - Compact VLM (< 4B params)

Evaluation Criteria:

- ✓ Open source availability
- ✓ Relevance to Physical AI use cases
- ✓ Hardware compatibility (ROCm/NPU)
- ✓ Community adoption & maintenance

Success Metrics & KPIs

Measuring Lifecycle Effectiveness

Model Lifecycle KPIs:

Metric	Target	Current	Trend
Average Time to Enablement	< 2 weeks	2.5 weeks	⬇️ Improving
Models in Profiling Stage	5-8 concurrent	6	➡️ Stable
Benchmark Pass Rate (1st try)	> 80%	75%	⬆️ Improving
Fine-tuning Speedup (avg)	> 2x	2.3x	⬆️ Good
CI/CD Pipeline Success Rate	> 95%	97%	✅ Excellent

Portfolio Metrics: Total: **45+** | Production-Ready: **28** | In Development: **12** | Experimental: **5**

Roadmap & Next Steps

2026 Execution Plan

Q1 2026 (Current):

-  Establish lifecycle framework
-  Complete 8 models through full lifecycle
-  Deploy GitHub Pages documentation

Q2 2026:

-  Expand to 15 production-ready models
-  Implement automated benchmarking CI/CD
-  Onboard 2 external partner teams (MCW/SOW)

Q3 2026:

-  Multi-model orchestration (5 system pipelines)
-  Add 10 emerging models from watchlist
-  Scale fine-tuning operations

Q4 2026:

-  Full production deployment for 3 customer segments
-  Documentation portal v2.0
-  2027 strategy planning

Call to Action

Building the Future of Physical AI

Our Commitment:

🎯 **Systematic** approach to model support | 🎯 **Efficient** collaboration with partners
🎯 **Rapid** adaptation to emerging technologies | 🎯 **Quality** focus through CI/CD

Get Involved:

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Together, we enable Physical AI at scale.